

Academic Thesis for the Attainment of the Degree of
Master of Science
at the Technical University of Munich

An Algorithmic Framework for Very Large-Scale Capacitated Vehicle Routing Problems

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An Abstract should not include more than 150 words.

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Abbreviations

CVRP Capacitated Vehicle Routing Problem.

1 Introduction

Motivation: Whats the real world problem and application, why is it important? Why are we doing this research now? Mention the BKS challenge and the project study as underlying motivators for this thesis.

Status Quo: What is the current state of the art? What are the limitations of the current state of the art? What are the challenges? Mention np-hardness and the issues that come with monolithic approaches at scale.

Research Questions: What is the focus of this thesis? Name primary and secondary research question and outline how they will be answered/evaluated.

Contributions: Specify the concrete contributions made by this thesis. This should include the general approach, and specifically the novel contributions of HAOS, BG-AILS, and the parallelized DRISPI framework approach.

Structure: Overview of the chapters and their content.

2 Literature Review

Cite papers with `\cite{BibTeX key}`

There is a comprehensive introduction to Operations Research by Winston (2007). It may be beneficial to consult comprehensive review articles within the relevant research domains (Dekker et al., 2012; Matl et al., 2018; Zhang and van Woensel, 2023).

Also use the introduction to this chapter to outline the general field of VRP and related problems. Explain np-hardness, the background VRP and CVRP, and how research has been progressing in the field.

2.1 State-of-the-Art Solvers for Large-Scale CVRP

Namedrop the big monolithic solvers for CVRP and their limitations. Especially mention the four solvers which are used in DRISPI: AILS-II, FILO, FILO2, and HGS-CVRP.

The main limitation to be mentioned here is the computational complexity and how it scales superexponentially with size of the problem.

Also explain the general approaches of these monoliths and what else is out there.

2.2 Scalability Strategies

Write here about different approaches to scale when it comes VRPs. This should include parallelization (-> prepares for DRISPI), decomposition or divide and conquer approaches (-> prepares for DRI), and maybe some recent machine learning approaches (-> prepares for HAOS).

2.2.1 Parallelization

xyz

2.2.2 Set Covering and Set Partitioning

xyz

2.2.3 Decomposition

xyz

2.2.4 Machine Learning

xyz

2.3 Algorithmic Frameworks

Idk about this section, but this would be a great way to introduce the approaches of Christoph (DRI for CCRPTW and DRSCI for HVRP). Maybe there are other approaches like this which include different strategies.

I need to be clear here what the difference between a framework, solver, and heuristic is. But if done well, this is a very nice way to introduce these terms and concepts and a good bridge to the research gap.

2.4 Remaining Research Gap

This should seamlessly connect to the research questions and pick them back up. It should present the state of the art approaches as unfit for the future of larger and larger VRPs.

Also, it should present that a combination of different strategies to tackle scale is unexplored and that the release of this new problem set is the perfect opportunity to explore this.

This section should get a sick figure/table to illustrate the research gap and how DRISPI combines all these scalability strategies which have been explored only in isolation so far.

3 Problem Statement

As introduced in Chapter 1, this work addresses the Capacitated Vehicle Routing Problem (CVRP) at a very-large scale. We now state the problem formally, discuss its computational complexity, and define the scope of the computational study.

The CVRP is defined on a complete undirected graph $G = (V, E)$, where $V = \{1\} \cup C$ is the set of locations. Node 1 represents the depot, and $C = \{2, \dots, n+1\}$ is the set of customers to be served, so that $n = |C|$. Each edge $\{i, j\} \in E$ carries a non-negative travel cost $d_{ij} \geq 0$, and each customer $i \in C$ has an associated demand $q_i > 0$. A homogeneous fleet of vehicles, each with a maximum capacity Q , is dispatched from and returns to the depot. The objective is to find a set of routes, each starting and ending at the depot, such that every customer is visited exactly once, no route exceeds the vehicle capacity, and the total travel cost is minimized.

The CVRP is NP-hard (?), a direct consequence of its generalization of the Traveling Salesman Problem. In practical terms, the number of feasible solutions grows factorially with n , rendering exact methods intractable beyond a few hundred customers. While branch-cut-and-price algorithms can solve small and medium instances to proven optimality (?), they are not viable at the scales considered here. This necessitates heuristic and metaheuristic approaches, which trade optimality guarantees for high-quality solutions within practical time budgets. Even within the heuristic paradigm, as Chapter 2 discusses, the cost of operating on the full problem simultaneously grows rapidly with n : neighborhood evaluation, memory requirements, and local search effectiveness all deteriorate at very-large scale, motivating the decomposition-based approach pursued in this work.

This work focuses specifically on the XL instances of the CVRPLib benchmark set (?), containing between 1,000 and 10,000 customers. These instances span a wide range of structural characteristics, including depot positioning, customer distribution, demand heterogeneity, and average route length, making them a demanding and representative testing set for large-scale CVRP solvers. DRISPI, proposed in Chapter 4, is designed specifically to operate at this scale by decomposing the full instance into tractable subproblems, each solved independently using existing state-of-the-art solvers.

4 Framework

General introduction to the framework. Build a figure to show how the framework would work sequentially here maybe (since this is easier to understand). It is just a short introduction.

The first part should also drop the name DRISPI for the first time and explain its components in one paragraph.

Pseudocode and a figure of the finally used framework architecture should come at the end of this chapter, since it won't be understandable at this point.

Instead, explain the main ideas behind the framework and namedrop important aspects for the later chapters (such as the route pool).

Maybe some of the following sections deserve subsection, but I am finding it quite clean to have it all on the same hierarchy level.

4.1 Hierarchical Adaptive Operator Selection (HAOS)

Explain the general approach of HAOS. Explain the online learning and make the connection to the bandit and ML literature.

Outline the different levels and the reward system. Maybe already mention warm-up periods, etc.

4.2 Decomposition

This has to outline route-based and vertex-based clustering. And it has to outline the different methods used within each of these paragraphs.

This also has to include dissimilarity measures and the impact of demand in addition to spatial proximity.

Some nice figures would be helpful here (maybe show the impact of how clustering differs on the same instance with fixed HAOS operators (except for levels 2-4)).

4.3 Routing

Explain how the subclusters are solved. Quickly mention parallelization of this stage. Also include recombination and compare with other decompose and solve parallel approaches.

Also explain the hyperparameters for this stage. Should be quite short (1-2 pages at most).

4.4 Boundary-Guided Adaptive Iterated Local Search

Explain the logic of BG AILS. Show the important parameters and some nice plots to illustrate the approach.

I want one plot to explain route selection and perturbation.

And I want one plot that shows the difference between BG-AILS and AILS. (Highlight the nodes that were touched, intensify color by number of touches)

Since this is one of my novelties, it can be a longer section.

4.5 Set Partitioning

Explain SP and the SC fallback. Give the formal problem statement of SP and its LP Relaxation. Do the same for SC, but maybe in the appendix.

Explain the role of SP and min coverage.

4.6 Improvement

This section can be super short. It should just explain the purpose.

4.7 Pipeline Orchestration

This section should lead all previous sections together.

Here I need to explain with a figure how the pipeline is parallelized (the 6+1+1 approach) to minimize idle cores.

I also need to write a long pseudocode explaining the entire pipeline logic. This could be hard if done in the parallelized setting. Will decide how to do this later.

5 Computational Study

xyz?

5.1 Experimental Design

xyz?

5.1.1 Framework Implementation

Write which hardware is used and which external components/implementations are used in the utilized framework (Solvers, Gurobi, etc.).

5.1.2 Dataset

Mention the new XL dataset, how it is generated and composed (the different levers n , k , dep , customer distribution, demand distribution, Q , r).

Also explain how they were initially evaluated and what existing solutions are used for the later benchmarking (AILS-II, FILO, FILO2, and HGS-CVRP, maybe DRSCI). Explain how current BKS were found through the BKS challenge and that they serve as the baseline instead of the numbers from the original paper. (Provide link and date to the official website from which the BKS values were obtained before the final benchmark).

5.1.3 Repititions

Explain the number of seeds used and why. Reference Law and Kelton here and show the statistics behind why this is clean and answers the research question properly.

Name of the chapter is object to change.

5.1.4 Hyperparameters

Show the hyperparameters used for the benchmark run. Explain the impact of the different hyperparameters on the performance.

5.2 Results

Short introduction.

5.2.1 Performance Comparison against State-of-the-Art Monolithic Solvers

Show the results of the 100 runs a 2h with 3 seeds each. Make a big table with the avg. and best results for each instance. Next to it take the values from the Queiroga et al. (2026) paper and show the difference.

This answers the primary research question.

5.2.2 Performance by Instance Characteristics

Show how DRISPI performs by instance characteristics. This should be a simple plot showing the performance by instance size, depot position, customer distribution, and demand distribution.

5.2.3 HAOS Weights

Investigate how HAOS weights are converging and if there are obvious tendencies. Interpret the results.

Are some HAOS levels converging always to the same result? Are some HAOS levels always converging in a similar way subject to the instance characteristics? Whats the variance in HAOS weight convergence? Do they differ a lot from seed to seed and are sensitive to randomness?

Are the results interpretable in general?

5.2.4 Convergence Behavior

Show how the convergence behavior is like for DRISPI by wallclock time and iterations. Use this chapter to also investigate improvement sources (quantitatively and qualitatively).

5.2.5 Optional: Long-Run Study

Another study to show the performance of DRISPI with more time budget. This should focus especially on instances with solutions close to the BKS.

5.2.6 Optional: Scalability Study

Another study to show the diminishing returns of the scalability through parallelization. Use the same approach as in the scalability chapter of the literature.

5.3 Limitations

Lay out the limitations of the study. It should include the difficulty of comparing single-threaded monolithic solvers to the parallelized framework (CPU time vs. wallclock time, different hardware, etc.).

Also mention limitations and variations during the run when it comes to hardware and number of seeds. Computation was a limiting factor here and definitely influenced performance.

6 Conclusion

Short introduction. It should reflect on the primary and secondary research questions.

6.1 Summary

Mention the strengths (such as adaptiveness through online learning) and weaknesses (such as more CPU time required) of the framework.

Also elaborate on the main contributions of the thesis.

6.2 Future Research

Use the HAOS weights to train an offline learning model and inject the results back into the framework. This effectively eliminates the warm-up period and will lead to a substantial speed-up of the framework.

Also mention that the framework, other than monolithic solvers, is highly adaptable to different instances and VRP variants. Since it only requires the plug-and-play of capable clustering methods and solvers on their respective stages.

Bibliography

- Dekker, R., J. Bloemhof, I. Mallidis. 2012. Operations research for green logistics – an overview of aspects, issues, contributions and challenges. *European Journal of Operational Research*, 219 (3), 671-679, doi: 10.1016/j.ejor.2011.11.010.
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- Winston, W. L. 2007. *Operations research: Applications and algorithms*. 4th ed. Brooks/Cole Thomson Learning, Belmont 2007.
- Zhang, J., T. van Woensel. 2023. Dynamic vehicle routing with random requests: A literature review. *International Journal of Production Economics*, 256 108751, doi: 10.1016/j.ijpe.2022.108751.

Appendix

1 Appendix One

2 Appendix Two

Declaration Of Authorship

I hereby solemnly declare that I have completed the present work independently. All ideas taken directly or indirectly from external sources have been clearly identified as such.

I am aware that the work may be examined in digital form to determine whether unauthorized aids were used and whether it constitutes - either in whole or in part - plagiarism. For the purpose of comparison with existing sources, my work may be entered into a database and, after review, may remain there for comparison with future submissions. No further reproduction or usage rights are thereby granted. This work has neither been submitted to another examination authority nor published.

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Maximilian Mustermann

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